

Muhammad Ali

AI/ML RESEARCHER · SOFTWARE ENGINEER

Skardu, Gilgit-Baltistan, Pakistan

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Summary

AI/ML researcher and Software Engineering graduate with hands-on experience in natural language processing, low-resource speech recognition, and retrieval-augmented generation (RAG). Proven ability to design, evaluate, and deploy end-to-end AI systems, including transformer-based ASR and hybrid semantic search pipelines. Passionate about building language technology for underrepresented communities and advancing research in Large Language Models (LLMs) and multilingual NLP. Comfortable across the full research-to-deployment cycle, from corpus construction to production deployment.

Education

The Islamia University of Bahawalpur (IUB)

Bahawalpur, Pakistan

B.S. IN SOFTWARE ENGINEERING

Mar. 2022 - Feb. 2026

- Final Grade: CGPA 3.62/4.00
- Recipient of the Alkauser Merit-Based Scholarship (MAWA Educational Support Program) for sustained academic excellence (CGPA $\geq$ 3.5).
- Relevant Coursework: Artificial Intelligence, Machine Learning, NLP, Deep Learning, Data Structures & Algorithms, Linear Algebra, Probability & Statistics.
- Medium of Instruction: English

Skills

Programming Languages	Python (Advanced), C/C++ (Proficient), C#, JavaScript
Frameworks & Libraries	PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, SciPy, spaCy, Hugging Face, Sentence Transformers
Tools & Platforms	Git/GitHub, Google Colab, Jupyter Notebook, VS Code, SQL Server, MySQL, Firebase
Soft Skills	Team Collaboration, Problem-Solving, Time Management, Adaptability, Communication
Languages	Urdu & Balti (Native), English (Professional Working Proficiency)

Research & Technical Projects

BaltiVoice: Speech Corpus and Whisper-based ASR System for Balti Language

AI/ML RESEARCH PROJECT

May 2026 – Jun 2026

- Developed the first publicly available Automatic Speech Recognition (ASR) system for Balti (ISO 639-3: bft), a low-resource Tibetic language.
- Created and released a 16.8-hour speech corpus containing 10,060 validated utterances from 136 speakers on Hugging Face under CC0 license.
- Fine-tuned OpenAI Whisper-small achieving 26.74% WER and 8.67% CER on a speaker-disjoint validation set.
- Deployed an interactive transcription demo with microphone and file upload support using Hugging Face Spaces.
- Technologies: Python, PyTorch, Hugging Face Transformers, OpenAI Whisper, Speech Processing.

AI-Based Quranic Verse Search Engine with RAG

FINAL YEAR PROJECT

May 2025 – Feb 2026

- Designed a hybrid retrieval system combining TF-IDF keyword matching and Sentence Transformer semantic embeddings over a 6,200+ verse corpus.
- Developed a Retrieval-Augmented Generation (RAG) pipeline using LangChain for context-aware and grounded responses.
- Optimized vector embedding caching to reduce search latency and improve scalability under concurrent usage.
- Built a responsive web interface with Tafsir retrieval, audio playback, and dynamic verse card generation.
- Technologies: Python, PyTorch, Sentence Transformers, LangChain, TF-IDF, Tailwind CSS, AJAX.

Document Intelligence Assistant using Retrieval-Augmented Generation

GENERATIVE AI PROJECT

2026

- Built an AI assistant capable of answering questions from user-provided documents using semantic retrieval and LLM-based generation.
- Implemented document processing, embedding generation, vector search, and response generation pipeline.
- Technologies: Python, LangChain, LLMs, Vector Database, NLP.

Experience

Enigmatix (Pvt) Ltd.

Bahawalpur, Pakistan

AI AND ML ENGINEER INTERN

Jan. 2025 - Jun. 2025

- Worked as an AI and ML Engineer Intern, gaining hands-on exposure to real-world machine learning pipelines and software development activities.
- Demonstrated strong technical knowledge and work capacity in applying AI/ML concepts to practical engineering challenges.
- Collaborated with the engineering team to develop and test models, showcasing a fast learning curve and passion for hard work.

- Completed a intensive virtual internship program focused on Artificial Intelligence, demonstrating dedication and consistency across collaborative tasks.
- Engaged in hands-on problem-solving projects involving real-world AI applications and data-driven decision making.
- Developed practical skills in AI engineering workflows while maintaining a growth mindset throughout the program duration.

Pre-prints & Publications

BaltiVoice: A Speech Corpus and Fine-tuned Whisper ASR System for the Balti Language

arXiv:2606.03504

FIRST AUTHOR

Jun. 2026

- Introduced the first publicly available speech corpus and ASR system for Balti, a low-resource Tibetic language.
- Released a 16.8-hour speech dataset and fine-tuned Whisper-small achieving 26.74% WER and 8.67% CER.

Research Interests

- Large Language Models (LLMs) and Generative AI
- Retrieval-Augmented Generation (RAG) and Knowledge-Enhanced NLP
- Multilingual and Cross-Lingual Natural Language Processing
- Low-Resource Language Technologies and Speech Recognition
- Deep Learning, Transformer Models, and Representation Learning

Extracurricular Activities & Volunteering

Gilgit Baltistan Student Council (GBSC), The Islamia University of Bahawalpur

Bahawalpur, Pakistan

VOLUNTEER COORDINATOR

Nov. 2021 – Present

- Led student orientation programs and coordinated communication for 100+ students from Gilgit-Baltistan.
- Organized cultural events promoting diversity, inclusion, and student engagement within the university community.

Alkhidmat Foundation Pakistan

Pakistan

STUDENT VOLUNTEER

2026

- Participated in community service activities focused on education support and social development.

Honors & Awards

Merit-Based Scholarship Recipient

2022–2026

Alkauser Merit-Based Scholarship, MAWA Educational Support Program

Certifications

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| • Machine Learning Specialization — Stanford University | Sept 2025 |
| • Deep Learning Specialization — DeepLearning.AI | Nov 2025 |
| • IBM Generative AI Engineering — IBM | Feb 2026 |
| • Generative AI with Large Language Models — DeepLearning.AI & AWS | Jun 2026 |
| • Mathematics for Machine Learning — Imperial College London | Nov 2025 |

Languages

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|---------|----------------------------------|
| Urdu | Native proficiency |
| Balti | Native proficiency |
| English | Professional working proficiency |

Recommendations

Dr. Muhammad Nauman

FYP Supervisor & Lecturer, Department of Software Engineering

The Islamia University of Bahawalpur

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Dr. Malik Muhammad Saad Missen

Associate Professor & Chairman, Department of Software Engineering

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